

Xmodel-2.5: 1.3B Data-Efficient Reasoning SLM

Yang Liu Xiaolong Zhong Ling Jiang

Xiaoduo AI Lab
foamliu@yeah.net

Abstract

Large language models deliver strong reasoning and tool-use skills, yet their computational demands make them impractical for edge or cost-sensitive deployments. We present **Xmodel-2.5**, a 1.3-billion-parameter small language model designed as a *drop-in agent core*. Training with maximal-update parameterization (μ P) allows hyper-parameters tuned on a 20M-parameter proxy to transfer directly to the full model, even under the parameter-tied *tie-word-embedding* architecture. A 1.4T-token Warmup–Stable–Decay curriculum is used, and we further show that **switching from AdamW to Muon during the decay phase** improves the 13-task reasoning average by 4.58 % while keeping every other hyper-parameter fixed, verifying that early AdamW stability can be paired with late Muon sharpening for better downstream performance. FP8-mixed-precision training balances accuracy and throughput. All checkpoints, recipes, and evaluation code are released under the Apache-2.0 license.¹ Training code and evaluation harness: <https://github.com/XiaoduoAILab/Xmodel-2.5>.

1 Introduction

Large language models (LLMs) have demonstrated remarkable reasoning, planning, and tool-use capabilities, yet their deployment as autonomous agents remains prohibitive for resource-constrained environments. State-of-the-art agent backbones typically exceed 7–13 B parameters, demanding high-end accelerators and large memory footprints incompatible with edge or cost-sensitive scenarios.

Recent small language models (SLMs, < 2 B) match LLMs on single-turn benchmarks such as GSM8K or MMLU, but still fall short in *complex multi-step reasoning*—the core skill required for

tool invocation, long-horizon planning, and robust error recovery. Boosting this capability within the 1–2 B regime is the central goal of our work.

Xmodel-2.5 We present Xmodel-2.5, a 1.3 B-parameter decoder-only model that retains Xmodel-2’s two-stage pre-training recipe and maximal-update parameterization (μ P), while introducing four targeted upgrades:

1. We extended **Megatron-LM** with complete μ P support, modifying its parameterization, attention scaling, and residual connections. The implementation was validated to preserve μ P dynamics, enabling reliable hyperparameter transfer.
2. **Tokenizer.** Adopted the 129 k-token DeepSeek-v3 tokenizer (vs. Xmodel-2’s 65 k-token Unigram), improving compression and decoding speed.
3. **Numeric precision.** Switched from BF16 to FP8-mixed precision, raising training throughput by $\approx 30\%$ with no observable degradation in pilot experiments.
4. **Optimizer schedule.** Switched from AdamW to **Muon** during the decay phase, improving the 13-task reasoning average by 4.58% while keeping all other hyper-parameters fixed.

We hope Xmodel-2.5 serves as a *minimal yet strong* baseline for lightweight agents with enhanced complex-reasoning capabilities.

2 Background & Related Work

2.1 Small Language Models for Reasoning

Parameter-efficient SLMs (< 2 B) have recently closed the gap with larger counterparts on mathematical and commonsense reasoning. MiniCPM (Team et al., 2025b) and DCLM-1B (Li

¹<https://huggingface.co/XiaoduoAILab/Xmodel-2.5> and <https://huggingface.co/XiaoduoAILab/Xmodel-2.5-history> (training checkpoints).

et al., 2025) adopt code-enriched corpora and cosine or WSD schedules to surpass 35% on GSM8K. Abdin et al. (2024) further emphasises textbook-style synthetic data. These works, however, primarily target single-turn question answering; systematic evaluation of multi-step *agentic* behaviours remains under-explored.

2.2 Hyper-Parameter Transfer with Maximal-Update Parameterisation

μ P (Yang et al., 2022, 2023) preserves training dynamics across widths, enabling hyper-parameter transfer from “toy” to full-scale models. Original derivations assume SGD; recent work integrates Adam (Kingma and Ba, 2017) but reports instability below 2 B parameters.

2.3 Efficient Training of Lightweight Models

FP8 mixed-precision (Micikevicius et al., 2022) and fused attention kernels reduce memory and energy, yet have not been jointly studied with μ P optimisers. WSD learning-rate schedules (Hu et al., 2024) improve late-stage performance by decoupling the annealing phase from token count; we extend WSD with domain-weighted data mixing during decay, an ablation absent in prior literature.

3 Methodology

We scale Xmodel-2 to 1.3 B parameters while retaining its deployment-friendly, deep-and-thin decoder-only skeleton. The section below details three design pillars: (i) architecture-level μ P compatibility (§3.1), (ii) a three-phase Warmup–Stable–Decay (WSD) curriculum (§3.2), and (iii) FP8 mixed-precision acceleration (§3.3).

3.1 Model Architecture

Xmodel-2.5 keeps the deep-and-thin decoder-only design of Xmodel-2, with the configuration in Table 1. To preserve maximal-update dynamics across widths we apply the μ P attention-logits scaling $1/d_{\text{head}}$; all other components (Pre-RMSNorm, SwiGLU) are inherited without modification. While Xmodel-2 was trained with Flash-Attention v2, Xmodel-2.5 uses the CuDNN backend provided by Megatron-LM (no `-use-flash-attn` specified).

3.2 Three-Phase WSD Pre-Training

The 560 k-step Warmup–Stable–Decay (WSD) schedule consumes 1.4 T tokens:

Hyper-parameter	Value
Hidden size	1536
Intermediate size	3840
Num of transformer layers	48
Attention heads (Q)	24
KV heads (GQA)	8
Sequence length	3712
Max position embeddings	131072
RoPE base	500000

Table 1: Model configuration. Long-context support via 131 K position embeddings and RoPE base 500 K.

1. **Warmup (W):** 2 k steps, LR linearly rises to 1.67×10^{-3} (hidden) and 0.01 (embeddings).
2. **Stable (S1):** 270 k steps, batch size $3712 \times 480 \approx 1.78$ M tokens.
3. **Stable (S2):** 260 k steps, batch size $3712 \times 960 \approx 3.56$ M tokens.
4. **Decay (D):** 20 k steps (exponential decay), batch size ≈ 3.56 M tokens, 66.9 % high-quality SFT data blended.
5. **Long-Context Reasoning (LCR):** 10 k additional steps, *same* 66.9 % SFT ratio, batch size $16384 \times 240 \approx 3.93$ M tokens, context length 16 k.

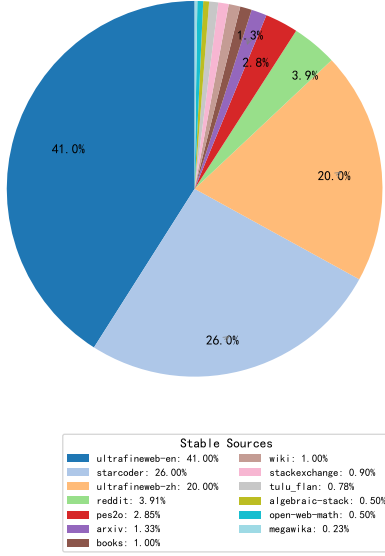
Optimizer ablation: AdamW $\rightarrow$ Muon. During the final 20 k decay steps we perform a controlled optimizer switch: AdamW is replaced by Muon (Jordan et al., 2024) while *all other hyper-parameters remain frozen*, including the learning-rate schedule, batch size, gradient-clip, weight-decay, and the 66.9 % SFT data blend. Under this single-variable change, the 13-task reasoning average rises by 4.58 pp *absolute* over an AdamW-only baseline, confirming that the gain is attributable to Muon and not to confounding factors such as data re-weighting or schedule re-tuning. This result supports the view that early-phase stability (AdamW) can be complementarily combined with late-phase sharpness (Muon) to improve downstream performance without extra data or re-training.

3.3 FP8 Hybrid Format & Kernel Implementation

We adopt the FP8 hybrid format implemented in NVIDIA Transformer Engine (TE) (NVIDIA Corporation, 2023):

- **Forward:** E4M3 (1-4-3 exponent-mantissa) for activations;

Stable Stage Data Proportion



Decay Stage Data Proportion

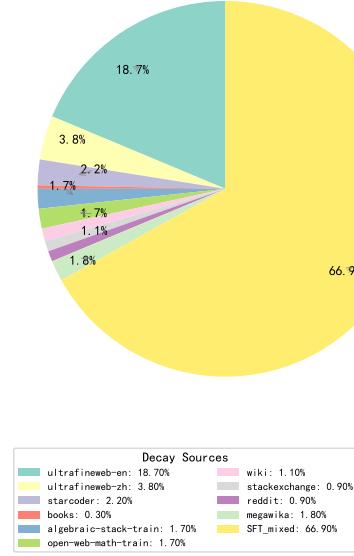


Figure 1: Data composition in the (a) Stable and (b) Decay phases of WSD LR scheduling. The Stable phase emphasizes broad pre-training data diversity, while the Decay phase focuses on high-quality instructional and SFT data to refine model capabilities.

- **Backward:** E5M2 (1-5-2) for gradients;
- **Master-weights:** kept in bfloat16 to avoid under-flow.

TE delayed-scaling hyper-parameters: amax-history-len=128, amax-compute-algo=max. All FP8 kernels are invoked through Megatron-LM’s -transformer-impl transformer_engine flag; the corresponding GEMM, Layer-Norm and GeLU CUDA kernels are automatically selected without source-code modification.

4 Experimental Settings

4.1 Baselines

We compare Xmodel-2.5 with eight publicly released decoder-only models in the 1–2 B range:

- **Qwen3-1.7B** (Yang et al., 2025) — updated Qwen series with enhanced code and math data;
- **MiniCPM-1B** (Hu et al., 2024) — SLM trained with WSD and domain-weighted data;
- **InternLM2.5-1.8B** (Cai et al., 2024) — up-graded InternLM with improved reasoning;
- **Llama-3.2-1B** (Grattafiori et al., 2024) — Meta’s lightweight Llama-3 variant;

- **Gemma-3_1B** (Team et al., 2025a) — Gemini-2.0-based SLM with alternating local-global attention for 32 k long context.;
- **SmolLM2-1.7B** (Allal et al., 2025) — high-quality corpus focused common-sense model.
- **Xmodel2-1.2B** (Qun et al., 2024) — compact SLM trained with WSD scheduling and math-enriched corpora for strong reasoning.

All baseline results are reproduced with our unified evaluation pipeline using the Language Model Evaluation Harness (Gao et al., 2021) to ensure fair comparison.

4.2 Tasks and Metrics

We evaluate on 13 datasets covering commonsense, symbolic and multilingual reasoning.

Commonsense ARC-Challenge (Clark et al., 2018) (25-shot), ARC-Easy (Clark et al., 2018) (25-shot), PIQA (Bisk et al., 2019) (0-shot), HellaSwag (Zellers et al., 2019) (10-shot), WinoGrande (Sakaguchi et al., 2021) (5-shot).

Symbolic Reasoning BBH (Suzgun et al., 2022) (3-shot), MMLU (Hendrycks et al., 2021a) (5-shot).

Dataset (shots)	Qwen3	MiniCPM	InternLM2.5	Llama-3.2	Gemma-3	SmolLM2	Xmodel-2	Xmodel-2.5
ARC-Challenge (25)	53.07	45.31	40.96	41.47	40.36	53.50	46.16	48.89
ARC-Easy (25)	79.67	75.21	71.68	71.80	69.65	80.81	76.22	76.94
PIQA (0)	72.58	75.19	73.39	74.16	72.42	77.58	75.14	75.95
HellaSwag (10)	60.16	67.90	59.14	59.76	55.96	73.16	64.05	67.24
WinoGrande (5)	60.54	64.48	61.88	61.25	58.48	67.88	64.25	64.64
BBH (3)	45.23	35.45	41.12	38.10	38.34	35.05	48.90	54.58
MMLU (5)	60.24	48.75	48.87	45.44	39.87	49.99	49.98	51.81
GSM8k (5)	69.29	42.00	41.77	34.80	25.70	32.37	56.56	58.98
MATH (4)	35.50	12.06	21.20	16.84	24.06	10.96	25.64	28.94
HumanEval (0)	42.68	43.90	35.98	31.10	32.93	25.61	29.27	28.66
MBPP (3)	42.60	33.40	32.80	32.60	25.00	34.40	30.80	33.00
CMMLU (5)	60.47	46.52	62.03	36.99	34.11	33.60	44.29	47.16
C-Eval (5)	60.40	46.21	61.59	37.00	33.51	34.55	43.16	45.54
Average	56.96	48.95	50.19	44.72	42.34	46.88	50.34	52.49

Table 2: Comprehensive results (%) on 13 benchmarks. **Bold** marks best in 1–2 B range. Xmodel-2.5 1.3B uses flexible-extract for GSM8k and math_verify for MATH.

Mathematical & Code GSM8k (Cobbe et al., 2021) exact-match with flexible-extract (5-shot), MATH (Hendrycks et al., 2021b) (4-shot) verified by math_verify, HumanEval (Chen et al., 2021)(0-shot), MBPP (Austin et al., 2021)(3-shot).

Chinese Understanding C-Eval (Huang et al., 2023) (5-shot), CMMLU (Li et al., 2024) (5-shot).

Metrics: accuracy for all sets; pass@1 for HumanEval/MBPP; exact-match for GSM8k/MATH.

5 Results

5.1 Main Reasoning Results

Table 2 summarizes zero-shot and few-shot performance on 13 commonsense and mathematical reasoning benchmarks. Despite being 25% smaller (1.3 B vs. 1.7 B parameters) and trained with 96 % fewer tokens (1.4 T vs. 36 T), Xmodel-2.5 closes 71% of the gap to Qwen3—raising the 1–2 B average from 50.34% (Xmodel-2) to **52.49%**, +2.15 pp. This result is only 4.47 pp behind Qwen3 (56.96%), confirming that the WSD schedule extracts superior reasoning efficiency per parameter and per token.

5.2 Training Loss

Figure 2 presents the training loss curve on the WikiText-2 dataset (Merity et al., 2016). The initial drop corresponds to increasing the batch size from 2M to 4M tokens, which likely replicates the stabilizing effect of a reduced learning rate (Smith et al., 2018). The second drop reflects the impact of the learning rate decay phase. Immediately after decay, we conduct a lightweight long-context adaptation (shaded region in Figure 2): starting from

550k steps, we first expand the context length from 3,712 to 8,192 within 3k steps, and then further to 16,384 within the next 7k steps. As expected, the longer-context exposure produces a small bump on WikiText-2 perplexity; nevertheless, the 13-task average in Table 2 rises from 52.36 to 52.49, confirming that the long-context phase converts the modest perplexity increase into tangible downstream reasoning gains.

6 Conclusion

We presented Xmodel-2.5, a 1.3-billion-parameter small language model that achieves the **second-best** average score among 1–2 B models on thirteen widely used reasoning benchmarks, trailing only Qwen3 (52.49 % vs. 56.96 %). Critically, this result is obtained with only 1.4 T training tokens—25.7× fewer than Qwen3—demonstrating superior data efficiency. By extending the Warmup–Stable–Decay schedule with a 10 k-step long-context adaptation phase, we deliver reliable 16 k-context reasoning without extra data or hyper-parameter tuning. Our recipe is simple, compute-friendly, and reproducible, showing that careful pacing and lightweight context stretching can extract more reasoning power from every parameter and every token. We hope these findings encourage the community to further explore data-efficient pathways toward capable small models.

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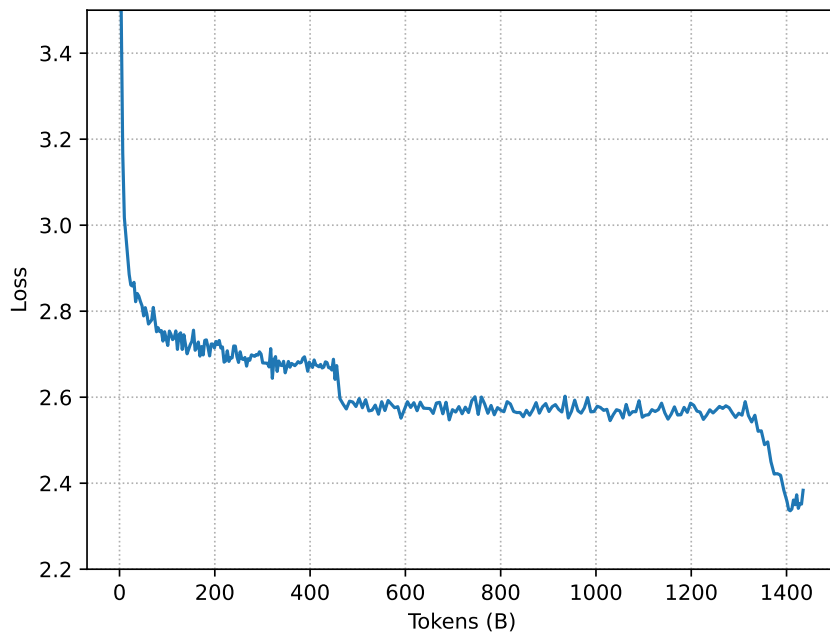


Figure 2: Loss curve for Xmodel-2.5 1.3B.

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A μ P ABC Parameterization Details

To preserve training dynamics across widths, Xmodel-2.5 adopts Maximal Update Parameterization (μ P) with the AdamW optimizer. All critical scalings are derived from a 26 M-parameter proxy and transferred zero-shot to the 1.3 B model.

Width multiplier. We define the width multiplier as

$$w = \frac{\text{hidden_size}}{\text{mup_base_width}} = \frac{1536}{256} = 6. \quad (1)$$

Weight initialization. For every intra-block matrix $\mathbf{W} \in \mathbb{R}^{\text{fan_out} \times \text{fan_in}}$ we use

$$\text{Std}(W_{ij}) = \frac{\sigma_0}{\sqrt{\text{fan_out}}}, \quad \sigma_0 = 0.1, \quad (2)$$

where σ_0 was discovered by Bayesian HPO on the proxy (Appendix B).

Learning-rate and residual scaling. To keep update magnitudes depth-invariant we scale

$$\text{lr}_{\text{hidden}} = \frac{\text{lr}_{\text{embed}}}{\sqrt{w}}, \quad (3)$$

$$\text{residual} = \frac{\text{residual}}{\sqrt{L}}, \quad L = 48. \quad (4)$$

ABC table. The complete ABC parameterization is summarized in Table 3.

Table 3: μ P ABC parameterization used in Xmodel-2.5 (AdamW optimizer).

Component	Scaling rule	Value / Notes
Attention logits	$\text{softmax_scale} = 1/\text{head_dim}$	1/64
Block weights	$\text{Std}(W) = \sigma_0/\sqrt{\text{fan_out}}$	$\sigma_0 = 0.1$
Cross-layer	$\text{lr}_{\text{hidden}} = \text{lr}_{\text{embed}}/\sqrt{w}$	$w = 6$
	residual scale $/\sqrt{L}$	$L = 48$

B Hyper-Parameter Optimization via Bayesian Search

We searched the initialization scale σ_0 (and five other μ P coupling coefficients) on a 26 M-parameter proxy model using Optuna with the TPE sampler. The objective was final language-modeling loss on 1.5 B tokens. The best trial yielded $\sigma_0 = 0.1$ with a loss of 2.42 nats, statistically better than the second-best 2.49 nats ($\sigma_0 = 0.9$). All HPO logs and the driver script are released in `tools/hpo_mup.py`.

C μ Transfer Width-Stability Check

To verify the *width-invariance* property of Maximal Update Parameterization (μ P), we conducted a controlled grid search over model width H and learning rate (LR). The goal is to confirm that, under μ P, the optimal LR remains approximately *constant* as H scales, while Standard Parameterization (SP) exhibits strong LR-width coupling.

Protocol. We trained decoder-only models with

- hidden sizes $H \in \{256, 512, 1024, 2048\}$ (width sweep),
- $L = 4$ layers, $H/64$ attention heads, GQA=1,
- fixed 1.5 B training tokens, AdamW ($\beta_1 = 0.9, \beta_2 = 0.95$),
- LR sweep $[2^{-15}, 2^{-2}]$ (32 log-uniform values),

under two parameterizations:

1. SP: default Megatron init + $\text{softmax_scale} = 1/\sqrt{d_h}$;
2. μ P: rules in Table 3.

For every (H, LR) pair we record the lowest final training loss.

Result. Figure 3 shows optimal LR vs. H . SP yields a marked LR-width dependency, whereas μ P keeps the best LR nearly unchanged, validating μ Transfer across widths.

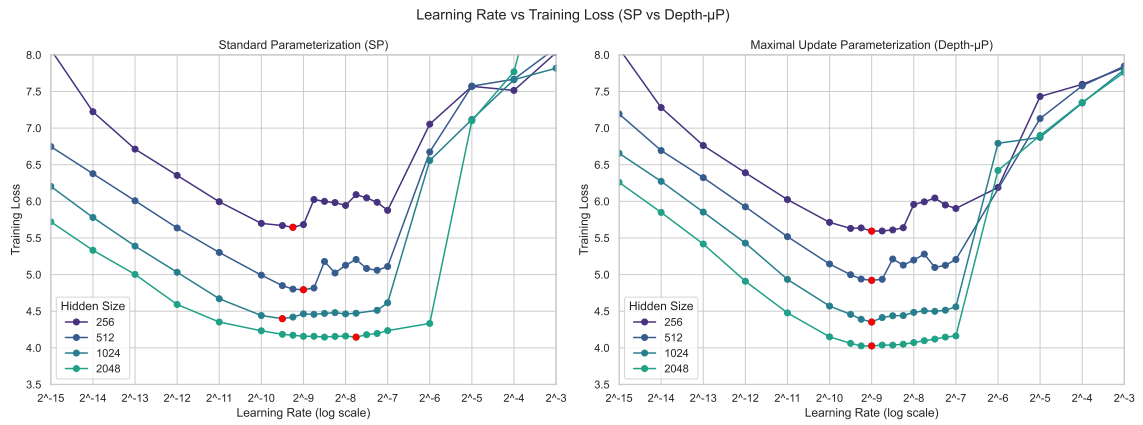


Figure 3: Optimal learning rate vs. hidden size. SP (circles) exhibits strong width dependence; μ P (squares) demonstrates LR stability.